

“THERMAL MAPPING STUDY PROTOCOL” FOR

[Company’s Logo]

- **Protocol Approval**

- *Protocol Prepared By: -*

Name : Eng.
Job Title : Technical Engineer
Sign. / Date :

- *Protocol Reviewed By: -*

Name : Dr.
Job Title : Quality Executive
Sign. / Date :

- *Customer Approval for Documentation of the following Protocol: -*

Name	Job Title	Sign./Date

- **Introduction**

- **ThermoCal Story & Its Sister Company ClimateSync™**



The company, with 20+ years in supply chain specialized in pharmaceutical and FMCG, by cooperation with Dr. ElFar, the founder of ClimateSync™, a Quality Pharmaceutical Scientist/Pharmacist, identified a significant gap in the **logistics** and especially **distribution** of pharmaceutical products in Egypt. This challenge particularly emerged in managing the **cold chain**, a temperature-controlled supply network crucial for maintaining the efficacy and safety of pharmaceuticals. The bottleneck arose from handling the numerous **thermal mapping studies** conducted by service providers associated with distributors. These studies ensure proper temperature control throughout the entire supply chain, from manufacturing to delivery. However, the current system for managing these studies was inefficient.

“As a quality and compliance pharmacist who is obsessed with technology disruption and innovation, I realized issues while working. I encountered difficulties in **manually entering vast amounts of data** from data loggers provided in the reports. These loggers record temperature fluctuations during transportation and storage. The process was not only time-consuming but also lacked transparency. As part of our quality review, we conduct operational reviews to **track** the time booked until the results are actually published and the end of the booked service to pay the service provider. Unfortunately, the current system for **retrieving past service requests** was very old-fashioned, time-consuming, and prone to losing data. Therefore, tracking the service provider's operation in handling our booked requests was challenging, making it hard to track the progress of studies in real-time before submitting the actual report”, Dr. ElFar said.

Problem(s) Identified

There were certain inefficiencies identified, which not only wasted valuable time and resources but also potentially jeopardized the quality and safety of time- and temperature-sensitive products due to potential delays in operation. These problems are as follows:

- Manual data entry of readings from service provider reports, creates a bottleneck, causing delays in reviewing and approving study reports.
- For security reasons and to comply with company policy, ISO-holder companies don't allow service providers or employees to bring in portable electronic devices, which made it harder to copy and paste data.
- Difficulty in tracking and managing booked studies due to reliance on hard copies.

- The company incurs extra costs due to a lack of real-time progress tracking.

Solution(s)

ThermoCal was born to address these challenges by:

- Digitizing the process through a secured portal with ready-to-use templates for reviewing and evaluating studies without manual data entry.
- Implementing an online booking system with automated creation of a validation masterplan for future services and easy access/retrievability of previous studies/services.
- Developing an application linked to study bookings to provide real-time updates on study progress, enhancing transparency.

Vision(s)

Short-term

Our goal is to provide our services through our digital platform to manufacturers and distributors in the pharmaceutical sector who handle time and temperature-sensitive products.

Medium-term

We aim to expand our services through our application/portal, offering real-time updates to all businesses handling time and temperature-sensitive products, including pharmaceuticals, non-pharmaceuticals, logistics, food and beverage, healthcare (test, research, and production laboratories)-related, and other relevant sectors.

Long-term

Venturing into manufacturing our own line of temperature monitoring and other service-related products, further solidifying us as a leader in the facilities industry and providing comprehensive solutions to our clients worldwide.

Mission(s)

We are here to empower quality control and assurance systems with efficient solutions to our valuable business partners.

Closing Remark(s)

ThermoCal solutions aim to streamline the review and documentation control processes of any quality department in sectors handling time and temperature-sensitive products. Our goal is to become the first digitized platform in the thermal mapping service sector in Egypt, positioning us ahead of our competitors. By offering real-time updates, saving labor time, ensuring cost-effectiveness, and promoting environmental sustainability through reduced paper usage. We aim to benefit any company that requests our services.

In this document, **ThermoCal** verifies [Insert the name of the place]'s conformance to specification in its intended operational environment.

- **Purpose**

This document sets out the qualification tests recommended by the engineering division to verify the conformity of [Insert the name of the place] to approved technical specifications. This document is also reviewed by our quality and operation divisions before being submitted officially to our clients.

The **data loggers (DLs)** used in all our services are of the same type [12] as listed in the table below, along with all specifications.

Manufacturer	-	
Type	Temperature and Relative Humidity Data Loggers	
Model	-	
Quantity	-	
Measurement ranges	°C	-30: 60
	% RH	20: 90
Area	- m ²	
Accuracy ranges	°C	± 0.5
	% RH	± 3
Memory	Capacity	- Measuring Values
Logging interval	- min	
Alarm limits	High	N/A
	Low	N/A
Calibration points	10°C and 30% RH.	
	20°C and 50% RH.	
	30°C and 70% RH.	

- **Overview**

- **Scope**

The protocol is designed to encompass thermal mapping and, if necessary, optional humidity mapping for [Name of the Place] during [Loading State] in [Enter Season].

- **Testing Philosophy**

The test plan aims to provide a reasonable level of assurance for every component and subsystem of the [Location] and [Floor Level], with particular

emphasis on **Critical Process Parameters (CPPs)** [9], [10]. While testing cannot be exhaustive, it endeavors to achieve comprehensive coverage.

CPPs are the key factors or variables that have a significant impact on the outcome or quality of a process. In the context of a thermal mapping study, CPPs typically include:

- **Temperature**
Monitoring and controlling temperature throughout the study area is crucial, as variations in temperature can affect the stability and quality of temperature-sensitive products.
- **Relative Humidity (RH)**
RH levels can influence the moisture content and stability of products, especially in environments where humidity control is essential.
- **Airflow**
Proper airflow is essential for maintaining uniform temperature distribution within the study area. Variations in airflow can lead to temperature gradients and affect the accuracy of the mapping results.
- **Time**
The duration of the mapping study is critical for capturing temperature fluctuations over an extended period, including variations due to seasonal changes or daily cycles.
- **Data Logger Placement**
The placement of DLs in the study area is crucial for capturing temperature and humidity readings accurately. Ensuring proper distribution and positioning of DLs is essential for obtaining reliable mapping results.
- **Calibration**
Proper calibration of DLs and monitoring equipment is essential to ensure accurate temperature and humidity measurements throughout the study.
- **Environmental Conditions**
External factors such as ambient temperature, sunlight exposure, and proximity to heating or cooling sources can influence temperature and humidity levels within the study area and should be considered during the mapping process.

By monitoring and controlling these CPPs, researchers and analysts can ensure the accuracy and reliability of thermal mapping study results, enabling them to make informed decisions about TTSP storage and handling.

All tests are crafted to validate the system against clear, documented, predetermined acceptance criteria. Protocols ensure that testing adheres to clear, predetermined methods approved in advance.

Tests are to be executed by properly authorized personnel, with support from relevant documentation and test equipment.

Whenever feasible, tests are structured to gather all necessary documentary evidence for retrospective audit of results. Procedures dictate that test results undergo formal review before being considered finally accepted.

As tests generate documentary evidence of results, they can be witnessed during the review stage. Whether the testing is independently witnessed at the time is at the discretion of the customer.

- **[Location] Identification**

Location	-
Area	m ²
A.C Setting	20°C
Acceptance criteria	Temperature 15:25°C

Responsibilities

The current paragraph delineates the responsibilities of the different entities involved in qualification activities.

- **ThermoCal Company**

ThermoCal Company will be responsible for the following activities:

- Writing and reviewing the thermal mapping study Protocol.
- Executing the thermal mapping and humidity mapping (if needed) study activities.
- Checking and collecting results into appropriate data collection forms.
- Writing and reviewing the final thermal mapping and humidity mapping (if needed) study.
- Summary reports that include the results of the study, including:
 - Raw data readings.
 - Calibration certificates of the DLs used in the study.
 - The final conclusion and recommendation.
 - Risk management.

- Conduct a site survey of the areas to be mapped.
- Gather information for each thermally separate area, including length, width, and height dimensions.
- Draw each area, depicting elements such as shelving or pallet racking that may affect temperature stability.
- Record the location of heating and cooling components, air distribution outlets, and temperature recording sensors.

STEP 4 – Establish acceptance criteria

- Define the required acceptance criteria in the protocol based on the type of temperature-sensitive products being stored.
- Clearly state the allowable “Temperature and Relative Humidity” limits within the mapped area.

STEP 5 – Determine DLs Locations

- Utilize the site survey findings to mark the necessary locations for the DLs.
- Provide the area layout, including length and width, for the thermal mapping study.
- Arrange the data loggers in a uniform pattern across all three dimensions—top to bottom, left to right, and front to back—to ensure adequate coverage of the area.
- Position the DLs every **5-10 meters** in width, distributing them to capture temperature gradients between the floor and ceiling in the warehouse.
- Consider the layout of the area, including factors such as squareness or alcoves, and how shelving and products may impact airflow.
- Ensure that DL positions align with the actual or planned storage locations of **TTSPs**.
 - For instance, it may be unnecessary to place DLs in areas like the upper part of high loading bays. Take into account other factors that may require adjustments to the number of DLs.
- Regarding height, arrange DLs vertically as follows:
 - If the ceiling height exceeds **3.6 meters**, organize DLs in vertical arrays at the bottom, middle (multiple), and top of the space. Designate:
 - **Location H:** High Level at **2 meters** Height.
 - **Location M:** Medium at **1-meter** Height.
 - **Location L:** Low Level at **0.3 meters** Level.
- The following **risks** should be considered as part of the mapping plan:
 - Storage areas near north-facing walls and windows may be affected by solar heat.
 - Lights and sunlight can serve as heat sources; potentially impacting goods placed on high racks in close proximity to a light source.

- Storage areas with long racks are likely to experience significant temperature and humidity variations from top to bottom.
- Consider the distance between the DL locations and return air grills, as positioning DLs to face return air grills can yield worst-case temperature readings.
- The density of the location can be represented by placing DLs in the middle.
- Presence of doors should be reflected in the distribution of DLs.
- Account for seasonal temperature changes or unusual weather events.
- Consider the volume of space inside the room.
- Ensure that devices are securely fastened to prevent damage or displacement during routine store operations.
- It may be appropriate to include a DL capable of monitoring door openings, programmed to align readings with door opening times.

STEP 6 – Record DLs locations

- Record the locations of DLs in a DL heights location table.
- Document the location identification and set point for each thermostat in the storage area.

STEP 7 – Label and program the DLs

- Label each DL with a unique ID from the DL heights location table.
- Program each device with a consistent recording interval, typically set to **10-15 minutes**
- Ensure all units have the same start time to enable time-correlated readings.
- Set the start time to allow for positioning of all units before recording begins.

STEP 8 – Fix DLs in position

- Securely fix each DL in position according to the DL heights location table and drawing.
- Ensure devices are positioned and fastened to prevent damage or displacement during routine operations.

STEP 9 – Conduct the mapping exercise

- Conduct the mapping study for either seven or three days As per the regulations, including the following steps:
 - Conduct the study under normal load conditions.

- At the study's conclusion, collect the DLs and verify their serial numbers and locations against the installation notes.

STEP 10 - Perform “power off” and “door/window” open tests

- **Door/Window Opening**
 - Open the first door/window for approximately **60 minutes as a worst-case scenario**, reflecting normal operational conditions since the warehouse will not typically remain open for such an extended period.
 - Note the time, close the door/window, and **wait for 30 minutes** to ensure that the DLs have reached the warehouse's conditions.
 - Open **one door/window** for approximately **60 minutes as a worst-case scenario**, in line with normal operation (If needed).
 - Open **all doors/windows** for approximately **60 minutes as a worst-case scenario**, in line with normal operation.
 - Note the time, close the doors/windows, and **wait for 30 minutes** to ensure that the DLs have achieved the warehouse conditions.
 - Analyse the temperature profile to evaluate the maximum allowable time for door opening and recovery time.
- **Power Failure**
 - Switch off the power supply for **60 minutes** as per normal operation.
 - Note the time and then switch on the power supply to the warehouse.
 - Wait for **30 minutes** based on previous experience to allow for the temperature to stabilize as the warehouse's conditions return to normal.

STEP 11 – Download and consolidate the data

- Download the DL readings and consolidate the data for the study analysis.
- Mark the areas of the hottest and coldest points in the warehouse.
- Review the report to identify the hottest and coolest points achieved, and calculate the **mean kinetic temperature**.

• Risk Management of Thermal Mapping Study

MKT reflects the cumulative thermal stress experienced by a TTSP and is commonly accepted for storage and excursion assessment during transit. However, its application must be justified by confirming that the stability-limiting characteristic of the product follows zero-order or first-order kinetics across the temperature range encountered.

When evaluating a temperature excursion using MKT, it's important not to rely solely on MKT. Additional information is essential, including the duration of the excursion, the excursion temperatures, whether temperatures exceeded 40°C, and the timeframe used to calculate MKT. At **TheroCal** & **ClimateSync™**, we adhere to these criteria.

Our software facilitates the calculation of MKT, and we identify hot and cold points by analyzing individual data values. In conclusion, our goal is to demonstrate scientifically, with documented justification, that the integrity and quality of the TTSP remain unaffected.

• Deviation Report

1. Deviation Summary Any temperature reading outside the approved range is recorded as a deviation, including abnormal behavior at the **Minimum**, **Maximum**, or **Most Fluctuated** sensor locations.

2. Critical Temperature Points

- **Minimum Point (Cold Spot):** Lowest recorded temperature.
- **Maximum Point (Hot Spot):** Highest recorded temperature.
- **Most Fluctuated Point:** Sensor with the largest temperature variation (ΔT).

3. Deviation Details

- Location
- Recorded value
- Expected range
- Duration
- Possible cause (door opening, equipment issue, power fluctuation, etc.)

4. Impact Assessment Deviation is classified as **Critical**, **Moderate**, or **Low Impact** based on risk to product quality and system performance.

5. CAPA (Corrective & Preventive Actions)

Corrective Actions:

- Adjust settings
- Repair/replace sensor
- Fix insulation/airflow
- Restore equipment performance

Preventive Actions:

- Improve door discipline
- Increase monitoring frequency
- Schedule preventive maintenance
- Staff training
- Revalidation if required

6. Closure Deviation is closed after implementing CAPA and verifying effectiveness, followed by Quality approval.

• Drawings

[Add Architectural Design Drawings]

- **Location H:** High Level at **2 m** Height.
- **Location M:** Medium at **1 m** Height.
- **Location L:** Low Level at **0.3 m** Level.

Note: One data logger should be located at the entrance door to monitor the environmental readings.

- **Attachments**
- *Raw Data and Chart of the Study.*
- *(Pre- and Post) Calibration Certificate of DLs.*
- *Environmental Reading Chart.*
- *Summary of Heat Distribution Study.*
- **Summary Report**

All study details, including the report and Excel files containing raw data, will be digitally accessible on the company's website portal. The summary report will be provided at the conclusion of the thermal mapping report. A hard copy of the study will be available upon request.

• References

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